

Module 2 Lab Exercise: Tools Used in Machine Learning

Individual Contribution

For this project, I completed my assigned portion of the Module 2 Lab Exercise independently. I worked through the Google Colab notebook and completed the required activities involving pandas, NumPy, matplotlib, and the Iris dataset.

I loaded and examined the Iris dataset and converted the dataset into a pandas DataFrame for analysis. I reviewed the structure of the dataset, including the number of samples, features, and classes. I also used pandas to calculate the mean measurements for each iris species and determine the number of samples belonging to each species.

I completed the NumPy exercise by calculating the mean and standard deviation of the sepal-length measurements. I verified the results using the notebook's built-in checks and confirmed that the calculations executed successfully.

I completed the data visualization exercises using matplotlib. I created a scatter plot comparing sepal length and sepal width by iris species and created a bar chart showing the number of samples for each species. I reviewed the visualizations to identify patterns and differences among the three classes.

I also completed the required analysis and reflection section of the notebook. I documented observations about the dataset, identified questions for further investigation, and reflected on what I learned about using pandas, NumPy, and matplotlib as part of a machine learning workflow.

After completing the notebook, I ran the cells to verify that the code executed successfully and that the required outputs and visualizations were displayed correctly. I also exported the completed notebook as a PDF.

Summary of Contribution

My individual contribution consisted of completing the Module 2 laboratory exercises, analyzing the Iris dataset, performing the required numerical and data operations, creating and interpreting visualizations, completing the required notebook documentation and reflection, testing the notebook, and exporting the completed notebook as a PDF.

The GitHub repository and upload of the completed notebook were handled by another group member and are therefore not included as part of my individual contribution.